would be wrong, or at least not advisable, to say therefore that a photon is a combination of an electron and a positron. For, conversely, a photon can be the product of an electron, for instance, when the electron jumps from one state to another. Moreover, a proton can change into a neutron and a positron, or a neutron into a proton and an electron. But one can hardly say that a proton is made up of a neutron and a positron. All these are true elementary particles, of which convertibility happens to be one of the characteristic properties.

In our survey of these elementary particles, we must look for properties capable of giving us a clue as to which ones are to be regarded as true building blocks of atomic nuclei, and which play a different part. For instance, the fact that a particle can emerge from the nucleus, is still not a proof of its being an ultimate, fundamental component part of the latter. This is evidenced by a simple consideration of the extra-nuclear structure of the atom. It consists of electrons. Yet, occasionally, also other particles emerge from it—namely: photons. But since these make an appearance only when certain changes in state of the extranuclear atomic structure occur, they are not referred to as integral component parts of that structure. Thus we make a distinction between particles which are always present in the extranuclear atomic structure—which we designate as its true component parts—and those which are produced within it, occasionally, due to changes in state, and depart from it subsequently. In the case of the extranuclear atomic structure, the former are the electrons, and the latter are the photons. In a certain sense, though, one might of course say that the photons were already present in that extranuclear structure. For although the space between the electron shells is empty (fast particles can penetrate it without any difficulty), yet it contains something—the electric field, which, to use an analogy, acts like the mortar that binds the building blocks of the extranuclear structure of the atom to the nucleus. In the wave aspect, the emitted light is an electromagnetic wave, the energy of which is drawn from that of this field. Fundamentally, it is merely a question of terminology whether we call this field a kind of substance or a property of space, and in this sense the photons may be said to have already been present as a field in the extranuclear structure. Nevertheless, it is helpful to make a distinction between the true building blocks of the nucleus and the field that holds them together, although this distinction cannot be of any fundamental importance. In any case, with respect to the extranuclear structure of the atom, there is a good reason for referring solely to electrons as its elementary building blocks and crediting the field with the capacity of occasionally producing photons. The following distinction may be made between the two kinds of particles: whenever the extranuclear atomic structure is subjected to outside interference—as, for instance, to bombardment by electrons or photons—the result is either that an